Week 16 — Extended Reading: Model-Based RL

STUDY-RL

October 6, 2026

1 Chua et al. (2018) — PETS

K. Chua, R. Calandra, R. McAllister, S. Levine. *Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models.* NeurIPS 2018, arXiv:1805.12114.
Summary. An ensemble of Gaussian dynamics models predicts $p(s_{t+1} \mid s_t, a_t)$; planning is by trajectory sampling under cross-entropy method (CEM). Matches SAC on several MuJoCo tasks with $\approx$ 5–10x fewer environment steps.
Why it matters. The clean MBRL baseline. Demonstrates that *uncertainty-aware* dynamics are necessary — a single deterministic model is exploited by the planner.

> "Probabilistic ensembles capture both epistemic and aleatoric uncertainty, and the planner naturally avoids regions where the models disagree."

2 Janner et al. (2019) — MBPO

M. Janner, J. Fu, M. Zhang, S. Levine. *When to Trust Your Model: Model-Based Policy Optimization.* NeurIPS 2019, arXiv:1906.08253.
Summary. Short-horizon ($H = 1$ to 25) branched rollouts from a learned model are concatenated with real data to train SAC. Theoretical bound relates model error and rollout length.
Why it matters. Shows that *short* model rollouts are the sweet spot: enough to amplify sample efficiency, short enough that error doesn't compound catastrophically.

> "We theoretically and empirically show that even a learned model with non-negligible prediction error can be used to improve a model-free policy by branching from real states and rolling out only a few model steps."

3 Hafner et al. (2023) — Dreamer V3

D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap. *Mastering Diverse Domains through World Models.* arXiv:2301.04104, 2023.
Summary. Single Dreamer agent with fixed hyperparameters solves Atari, Crafter, DMControl, Minecraft. Key tricks: symlog transformations on rewards/values, two-hot encoded value head, KL balancing, and discount weighting.
Why it matters. Demonstrates that latent world models scale, with stable training across very different domains – no per-task hyperparameter tuning.

> "The same agent solves over 150 tasks across continuous control, Atari, and Minecraft with a single set of hyperparameters."

4 Schrittwieser et al. (2020) — MuZero

J. Schrittwieser et al. *Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model.* Nature 588, 604–609 (2020).
Summary. MCTS over a *value-equivalent* learned latent model: no reconstruction, only value and policy targets. Matches AlphaZero on board games and AlphaGo on Atari without access to a simulator.
Why it matters. Shows that for downstream control purposes, the model only needs to be predictive of value – not visually faithful. A philosophical pivot for MBRL.

> "The model is trained end-to-end with the sole goal of accurately predicting quantities directly relevant to planning – the reward, the action policy, and the value function."

5 Alonso et al. (2024) — DIAMOND

E. Alonso, A. Jelley, V. Micheli, A. Kanervisto, A. Storkey, T. Pearce, F. Fleuret. *Diffusion for World Modeling: Visual Details Matter in Atari.* NeurIPS 2024 Spotlight, arXiv:2405.12399.
Summary. A diffusion model is the world model. Conditioned on a small action+history encoder, it produces highly faithful next-frame predictions. Policy is trained from rollouts of denoised frames; achieves strong Atari sample-efficiency at high visual fidelity.
Why it matters. Heralds the era of generative world models as a first-class policy training substrate.

> "Visual detail in the world model matters: the policy can lock onto features that a low-fidelity model averages away."

Sourced from `research/offline_modelbased_sota.md` (Track B, sections B.1–B.7).